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Unbounded Timing Terms Make a Composite Accident-Anticipation Score Video-Blind: Evidence from a Live Benchmark

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Abstract

Traffic-accident anticipation is increasingly scored by composite metrics that add an *earliness* term — time-to-accident at a fixed risk threshold — to conventional discrimination terms such as average precision and area under the ROC curve. We show that this composition has a failure mode that is structural rather than incidental: the discrimination terms are bounded on the unit interval, while the earliness terms scale with clip length, so a submission that maximises earliness can dominate the score while carrying no information about the video at all.

We demonstrate this on a live benchmark. On the *Zero-shot Accident Anticipation* competition (AUTOPILOT, 1,417 clips curated from MM-AU, 13 teams), a constant risk score of 0.51 — a submission that reads no pixels, has no parameters, and is one line of code — scores **2.35291** on the private leaderboard. That is identical to five decimal places to the eighth-placed team’s score, exceeds the scores of five of the thirteen teams including our own, and reaches 94.1% of the winning score. The entire advantage the first-placed entry holds over a curve that reads nothing is 0.14874, or 5.9% of its score.

We then recover the metric’s functional form from outside the competition. Using eighteen video-blind submissions as probes and differencing them, we isolate the score’s components without access to labels: the average-precision and area-under-the-curve terms contribute **0.35105** to a video-blind submission, while the two timing terms contribute **2.00186 — 5.70 times** as much. Score falls linearly in the frame at which risk crosses the threshold, at 0.016647 per frame, and this model predicts three held-out step-like curves to within 7×10^{-3} . It does not predict a linear ramp, which crosses at the same frame as a step yet scores 0.037 lower; we report that discrepancy rather than smooth it, since it bounds how far the crossing-frame model can be pushed. The same procedure recovers a property of the withheld ground truth: the accident onset lies between frames 100 and 140 in essentially every accident clip, with a mean of 120.3.

We report our own entry as the case study. It is a training-free ensemble of CLIP, a motion proxy and a text-anchor score; it placed ninth of thirteen with 2.00585, which is **below** the constant. We draw no conclusion about any other team’s method, and say so explicitly. We conclude with a six-point protocol whose central requirement is that every anticipation result be reported alongside the score of a video-blind control, and with the observation that an

earliness term must be bounded — by normalising to the per-clip maximum, or by being reported separately rather than summed — before it can be added to a discrimination term at all.

Index terms — accident anticipation, evaluation methodology, benchmark design, video-blind baselines, composite metrics, dashcam video.

1. Introduction

Road traffic injury remains among the leading causes of death worldwide, and the prospect of a system that sees a collision coming a second or two before it happens is what motivates the accident-anticipation literature. The field has grown quickly, and in the last three years it has acquired a second engine: vision-language models make it possible to build an anticipator without training on accident labels at all, which sidesteps the scarcity of annotated collisions that constrained earlier supervised work.

We set out to build such a system: a training-free ensemble that scores each frame of a dashcam clip for collision risk by combining CLIP similarity against danger and safety prompts, a motion-energy proxy, and a text-anchored score. We entered it in a public competition. It placed ninth of thirteen.

This paper is not about that system. It is about what we found when we tried to understand the score it received.

The competition, like a growing number of anticipation benchmarks, scores submissions with a *composite* metric: a weighted sum of average precision (AP), area under the ROC curve (AUC), and two earliness measures — time-to-accident (TTA) and stable time-to-accident (STTA), each evaluated at a risk threshold of 0.5. The composition is intuitive. A useful anticipator should both separate accidents from normal driving *and* raise the alarm early; summing terms that measure each seems a reasonable way to ask for both.

It is not. AP and AUC are bounded on $[0, 1]$. TTA and STTA are measured in frames and are bounded only by the clip length — 150 frames here. Summing a bounded quantity and an unbounded one means the unbounded one decides the ranking, and the arithmetic is not subtle: on this benchmark the timing terms are worth 5.70 times everything the discrimination terms contribute together.

Worse, the timing terms have a trivial global maximiser. TTA is the largest gap between the accident frame and any frame at which predicted risk exceeds the threshold; STTA is the largest such gap over which risk stays above the threshold continuously. A constant risk score of 0.51 crosses the threshold at frame 0 and never falls, so it attains, **on every clip, exactly the maximum value that clip admits** for both terms. No method that reads the video can beat it on those terms; a method can only match it. The remaining headroom is whatever AP and AUC are worth, and on this benchmark those two terms are worth 0.35105 in total — against 2.00186 for the timing terms that every submission can take for free.

So we submitted the constant. It scored 2.35291 against a winning score of 2.50165.

1.1 Contributions

1. **A measured demonstration on a live benchmark.** A video-blind constant ties the eighth-placed team's private score to five decimal places and outscores five of thirteen teams. Every *leaderboard score* in this paper was computed by the competition's own scoring service against ground truth we do not have; we implemented none of the metrics. All derived quantities — the decomposition, the fit, the ratios — are our arithmetic on those scores, and we mark them as such.
2. **Black-box decomposition of a composite metric.** We introduce a probe methodology — families of video-blind submissions designed so that unknown terms cancel under differencing — and use it to separate a metric whose weights are undisclosed and whose labels are withheld. We recover the AP+AUC floor, both timing terms, the score's slope in the crossing frame, and the support of the accident-onset distribution — and we report the one place the recovered model fails, which locates the residual structure it does not yet capture.
3. **A case study in which we are the subject.** Our own entry scores below the constant. We report this rather than omit it, because it is the clearest available illustration that a composite score of this shape does not measure what its name suggests.
4. **A corrected protocol**, six requirements traced to specific mechanisms, whose central item — report a video-blind control with every anticipation result — costs an afternoon and would have caught this.

1.2 What we do not claim

We wish to be explicit, because this paper reports a number that sits beside other teams' numbers.

We make no claim about the validity, quality or honesty of any other submission or published method. We did not re-run anyone's system, we do not have anyone's code, and we could not evaluate it if we did, because the labels are withheld from all entrants. Where we observe that the eighth-placed private score is numerically identical to the constant's, we are stating an arithmetic fact about two numbers on a public page and drawing exactly one inference from it: *the metric assigns that submission the same value it assigns a curve that reads nothing*. We do not infer what that submission contained. Several very different curves — any curve that exceeds 0.5 at frame 0 and never dips below it — receive that identical score, which is precisely the property under examination.

The claim of this paper is about a *metric*, demonstrated on one benchmark. It is not a claim about a field, and not a claim about anybody's method.

2. Background

2.1 The task

An anticipator receives a dashcam clip and emits a per-frame risk score $p_t \in [0, 1]$ for $t = 0 \dots T-1$. Frame indices are 0-based throughout this paper, because one frame is worth 0.0166 of score here and an off-by-one would exceed every residual we report. It is judged on whether it assigns high risk to clips containing a collision and low risk to normal driving, and on how far in advance of the collision it first does so.

2.2 Discrimination metrics

Treating each clip as a binary instance, average precision and area under the ROC curve are standard. Both are bounded on $[0, 1]$; AUC is 0.5 for a predictor that induces no ordering among clips, and AP equals the positive base rate.

2.3 Earliness metrics

Let t_{ai} be the annotated frame at which the accident begins, and t_a the first frame at which p_t exceeds a threshold τ . Then

$$TTA@{\tau} = \max \{ t_{ai} - t_a \mid p_t > \tau, 0 \leq t_a \leq t_{ai} \}$$

and, requiring the alarm to persist,

$$STTA@{\tau} = \max \{ t_{ai} - t_{a'} \mid p_t > \tau \text{ for all } t \in [t_{a'}, t_{ai}] \}.$$

Both are in frames (equivalently seconds, after dividing by the frame rate). Both are bounded above by t_{ai} , which is a property of the clip, not of the method. Both attain that bound when the risk score exceeds τ at frame 0 and never falls.

This is the fact the whole paper turns on. The maximiser of the earliness terms is a constant, and it is a constant regardless of what the clip contains.

2.4 The reference point, and why it is often ambiguous

t_{ai} requires an annotation. Where a corpus does not publish one — as here — entrants who wish to measure their own progress must substitute something, and the natural substitute is the end of the clip. Reporting $(T - t_a) / \text{fps}$ and calling it time-to-accident inflates every warning time by $(T - t_{ai}) / \text{fps}$, a quantity that depends on how the corpus was windowed and not at all on the method. We flag this because it is easy to do accidentally, and because we did it ourselves in an earlier draft of our own work.

3. The benchmark

Zero-shot Accident Anticipation, hosted on Kaggle by AUTOPILOT, ran from 17 February to 15 April 2026 and drew 59 entrants, 20 participants and 13 teams across 194 submissions. Late submission remains open and is scored against the same splits, which is how the experiments below were run. All competition figures in this section were read from the competition's own

Overview, Evaluation, Data and Leaderboard pages on 31 August 2026; archived copies accompany the submission.

Property	Value
Clips	1,417, a curated subset of MM-AU
Clip length	150 frames, 30 fps, 5.00 s, uniform
Modalities	RGB frames, driver gaze maps, text captions
Training split	none — inference only, by design
Published labels	none
Private leaderboard	≈70% of the test data
Submission	one JSON list of 150 risk values per clip

The evaluation page defines the score as

$$\text{score} = w_{\text{AP}} \cdot \text{AP} + w_{\text{AUC}} \cdot \text{AUC} + w_{\text{TTA}} \cdot \text{TTA@0.5} + w_{\text{STTA}} \cdot \text{STTA@0.5}$$

with the four weights “fixed by the organizers” and not disclosed. It specifies that positives are accident clips and negatives normal driving clips, gives the rank formulation of AUC, and gives the two timing definitions reproduced in §2.3 with t_{ai} explicitly the accident start frame *within the 150-frame clip*.

Three features of the benchmark matter for what follows.

No labels are published. `test.csv` carries `id`, `video_id`, `start_frame`, `end_frame`, `caption` and nothing else. AP, AUC, a false-positive rate, and any TTA measured to a true onset are therefore uncomputable by any entrant. This is not an oversight — it is what makes the competition a competition — but it has a consequence: entrants must develop against a locally computable proxy, and the proxy that is locally computable here is *the shape of one’s own risk curve*. A constant maximises every such proxy.

The organisers’ own reference submission is video-blind. The supplied `sample_submission.csv` contains a linear ramp from 0.001 to 0.999, identical for all 1,417 clips; it reads no pixels. That a blind ramp is the published starting point is unremarkable in itself; that it is not distinguishable *in kind* from the entries above it is the subject of this paper.

We note one thing we cannot yet resolve. The leaderboard carries a benchmark row scoring 0.58333, and it is tempting to attribute that score to `sample_submission.csv`. Nothing published says so, and our own ramp — which crosses the threshold at the same frame — scored 1.06725, so the attribution would contradict our own measurements. We therefore make no claim about what the benchmark row is, and have submitted the organisers’ file verbatim to settle it.

The captions describe outcomes. Seventy-nine distinct captions cover the 1,417 clips — “lead vehicle stops” (146 clips), “a vehicle controls loss” (144), “ego-car controls loss” (133). They state what happens, and they are supplied at inference time. Any method that conditions on them has access to the outcome it is being asked to anticipate. We note this as a second, independent

evaluation hazard in this corpus; our own system does not use them, and we did not probe it, so we make no measurement of its magnitude.

4. Method

4.1 The video-blind control family

A *video-blind* submission is one whose risk curve is a function of the frame index alone — identical for every clip, computed without opening a single frame. We use a family rather than a single curve, because a single flattering curve invites the objection that the strawman was tuned:

Family member	Definition
constant_0.51	$p_t = 0.51$ for all t
constant_0.99	$p_t = 0.99$ for all t
linear_ramp	p_t linear from 0.001 to 0.999
sigmoid_mu	logistic centred at frame 150μ , for $\mu \in \{0.40, 0.50, 0.60, 0.70\}$
step_at_k	$p_t = 0.49$ for $t < k$, 0.51 thereafter, $k \in \{0, 10, 25, 50, 75, 100, 125, 140\}$

4.2 Probes that isolate terms

The decomposition rests on one observation. **Every video-blind submission is identical across clips, so — provided the scorer reduces each 150-value curve to one clip-level score by any deterministic function of that curve — every submission in the family produces an all-way tie among clips, and therefore earns identical AP and identical AUC.** Whatever those terms are worth, they are worth the same to all of them, and they cancel exactly in any difference between two members of the family.

Two assumptions are doing work here and we state them rather than assume them. The first is that the reduction is clip-level at all: if AP and AUC were instead pooled over all $1,417 \times 150$ frames against per-frame labels, curve *shape* would enter them and the cancellation would fail. §5.3 reports a result that tests this, and §5.3 also reports one observation the assumption does not explain. The second is that the all-way tie is resolved conventionally (rank-averaging for AUC, giving 0.5; base rate for AP).

That turns the leaderboard into an instrument. Three probes suffice:

Probe	Curve	Isolates
never_crosses	0.49 throughout	the AP+AUC floor alone: both timing terms are zero
cross_then_drop	0.51 at frame 0, 0.49 after	floor + TTA (STTA is destroyed by the drop)
constant_0.51	0.51 throughout	floor + TTA + STTA

so that

- `floor = score(never_crosses)`
- `TTA term = score(cross_then_drop) - score(never_crosses)`
- `STTA term = score(constant_0.51) - score(cross_then_drop)`.

A fourth probe, `cross_then_dip` (0.51 throughout except a dip to 0.49 across frames 50–59), pushes the STTA reference point back to frame 60 while leaving TTA untouched, and provides a consistency check on the attribution. It is reported, with its result, in §5.2.

The `step_at_k` sweep measures the score’s dependence on the crossing frame directly, giving a second, independent route to the same quantity. Two routes to one number is what makes the result safe to publish.

The instrument also checks itself: `step_at_000` and `constant_0.51` are separately generated files describing the same curve, and they receive identical public and private scores. (`step_at_140` also scores identically to `never_crosses`, but we use that equality as a *measurement* in §5.4 and so cannot also count it as a validity check.) Throughout, “identical” means agreeing at the five decimal places the leaderboard displays, which is all the precision it exposes.

4.3 Reproducibility of the submission path

Before spending a submission slot we regenerated the organisers’ `sample_submission.csv` from its own parsed values and diffed it against the distributed file: byte-identical, 1,417 of 1,417 rows. This confirms the identifier ordering, the JSON list formatting and the float representation (Python `repr`) without consuming a slot, and it means a scoring anomaly cannot be attributed to our file writer. Every submission is validated for length 150, finiteness, and range `[0,1]` before upload.

4.4 Our own system, as the case study

The entry that placed ninth is a training-free ensemble of three per-frame signals: (i) CLIP ViT-L/14 similarity of each frame against a danger prompt and a safety prompt, converted to a risk score by softmax; (ii) a motion-energy proxy computed from successive frame differences; and (iii) a text-anchor score in which a sentence embedding of each of three fixed strings, selected by a motion threshold, is compared against sudden- and gradual-event anchors. The three curves are combined by fixed weights and smoothed. No component is trained on accident labels; the “zero-shot” claim is with respect to accident supervision, not with respect to the pretraining of CLIP or the sentence encoder.

It scored 2.00585 with two submissions and placed ninth of thirteen.

5. Results

All scores below were computed by the competition’s scoring service on the private split ($\approx 70\%$ of the test data), against labels no entrant holds.

5.1 A constant matches the eighth-placed score

The final private leaderboard, as published:

Rank	Team	Private score
1	CVLAB	2.50165
2	BUPT MIC Lab	2.46234
3	Tianhao Zhao	2.42361
4	cr-tfx	2.41555
5	wtiaw_tiw	2.39725
6	IsaacFI	2.39027
7	yyttl	2.36844
8	Paulini38	2.35291
9	SuryaInBytes (this work)	2.00585
10	Song_Ren	1.46008
11	WDL	1.35992
12	YooHyun.2	1.04203
13	Ratnachand Kancharla	1.03913
—	benchmark row (submission.csv)	0.58333

Our video-blind controls, submitted after the deadline and scored by the same service against the same private split, are *not* leaderboard entries and are reported separately:

Video-blind control	Private score
constant_0.51	2.35291
linear_ramp	1.06725
never_crosses	0.35105

The constant’s score and the eighth-placed team’s score agree at the precision the leaderboard displays. Per §1.2, the single inference we draw is that the metric assigns those two submissions the same value; we do not infer anything about what the team submitted, and we note that a whole family of curves — every curve exceeding 0.5 at frame 0 that never dips — receives that same value.

Three readings of these tables, in increasing order of importance.

The weak reading. A submission with no parameters and no input outscores five of thirteen teams.

The stronger reading. The gap between first place and the video-blind constant is **0.14874**. Whatever the winning system extracts from 1,417 clips, 150 frames each, three modalities, the metric values it at 5.9% of the score it awards that system. We quote the difference first and the

percentage second on purpose: the metric is a weighted sum with no meaningful origin, so a ratio of two scores can be made to say almost anything by shifting the scale. The difference and the decomposition below are scale-free; the percentage is a convenience, not evidence.

The strongest reading, and the one that generalises. As shown in §2.3, the constant is not a good baseline that happens to score well — it is the *exact maximiser* of both timing terms, simultaneously, on every clip. No submission can exceed it on those terms; a submission can only match it, and then compete on AP and AUC. Every point of separation visible on this leaderboard above 2.35291 must therefore have come from the discrimination terms, and §5.2 shows those terms are worth 0.35105 to a blind submission. The 0.14874 that separates first place from a constant is what the discrimination terms bought the winner above chance.

5.2 The decomposition

Probe	Private score
never_crosses	0.35105
cross_then_drop	1.22698
cross_then_dip	1.85347
constant_0.51	2.35291
constant_0.99	2.35291

Differencing as in §4.2:

Component	Contribution
AP + AUC, to a blind submission	0.35105
TTA@0.5, saturated	0.87593
STTA@0.5, saturated	1.12593
Timing total, saturated	2.00186 = 5.70 × the discrimination floor

Read that ratio precisely. It compares the timing terms *at their maximum* against the discrimination terms *at chance* — not maximum against maximum. The maxima ratio is not identifiable from outside, because $w_{AP} + w_{AUC}$ is undisclosed; but it is bounded, because the winner's discrimination terms are worth at least $0.14874 + 0.35105 = 0.49979$, so $(w_{TTA} + w_{STTA}) \cdot E[t_{ai}] / (w_{AP} + w_{AUC}) \leq 2.00186 / 0.49979 = 4.01$. Both framings say the same thing and we use the measured one: **a submission that reads nothing collects 2.00186, and the best discrimination anyone demonstrated on this benchmark added 0.14874 to it.**

The composite is not a weighted average of four comparable quantities. It is a timing score with a discrimination correction.

Note also that constant_0.51 and constant_0.99 score **identically**. The metric reads only whether the curve is above 0.5; it never reads by how much. A system that is certain and a system that is barely committed receive the same credit, which removes any incentive to calibrate.

The consistency check, and its residual. `cross_then_dip` dips below the threshold across frames 50–59, which should move the STTA reference point to frame 60 while leaving TTA at frame 0. With the per-frame STTA coefficient implied above ($1.12593/120.1 = 0.009375$), that predicts $2.35291 - 60 \times 0.009375 = 1.79041$. The measured score is **1.85347** — the observed loss corresponds to a reference point at frame 53.3 rather than 60, an 11% discrepancy. So the additive TTA/STTA attribution is close but not exact. The floor and the timing total are measured directly from flat curves and are unaffected; it is the *split* of 2.00186 into 0.87593 and 1.12593 that this residual puts a tolerance on, and we report the split with that caveat attached rather than to five decimals of implied precision.

5.3 The score is linear in the crossing frame

Crossing frame k	0	10	25	50	75	100	125	140
Private score	2.35291	2.18624	1.93732	1.52088	1.10445	0.68822	0.37967	0.35105

Over $k \leq 100$ the relationship is a straight line:

$\text{score} = 2.35304 - 0.016647 k$, i.e. **one point of score per 60.1 frames of delay**, or 0.4994 per second at 30 fps.

Past $k = 100$ the curve bends, as clip after clip has its accident occur before the alarm fires, and by $k = 140$ the score has returned to the floor.

Two conventions have to be fixed before any curve's crossing frame can be stated, and both were determined empirically rather than assumed: indices are 0-based, and the comparison is strict ($p_t > 0.5$). We report the crossing frame of every submitted curve as the index of its first value exceeding 0.5, computed from the submitted file itself. One frame is worth 0.0166 of score, so an off-by-one here would exceed every residual below.

Held-out prediction. Five curves were submitted independently and were not used to fit the line:

Curve	Crosses at	Predicted	Measured	Error
sigmoid_m0.40	60	1.35422	1.35428	+0.00006
sigmoid_m0.50	75	1.10452	1.10435	-0.00017
sigmoid_m0.60	90	0.85481	0.84840	-0.00641
linear_ram p	75	1.10452	1.06725	-0.03727
sigmoid_m0.70	105	(0.60511)	0.55763	(-0.04748)

The first three are predicted to within 6.4×10^{-3} , and the best of them to 6×10^{-5} . `sigmoid_m0.70` crosses at frame 105, beyond the $k \leq 100$ region where the line was fitted

and where the curve is already bending, so the model is not expected to hold there and its residual is bracketed. We list it because omitting a held-out point that missed would be the wrong kind of reporting.

Linear_ramp is a genuine discrepancy and we do not have an explanation. It crosses at frame 75, exactly like sigmoid_m0.50 (1.10435) and step_at_075 (1.10445) and a third curve, step_at_half (1.10442) — three independent curves agreeing to 1×10^{-4} — yet it scores 0.0372 lower. That gap is 2.24 frames of slope, which is not an integer, so it cannot be a crossing-frame effect. Two candidate explanations survive, and each costs something:

- The scorer's AP or AUC is *not* invariant to curve shape among video-blind submissions — for instance because it pools frames rather than reducing each clip to one value. This would undermine the cancellation argument of §4.2. It is, however, hard to reconcile with step_at_140 and never_crosses scoring identically, since those two curves have very different frame orderings.
- The timing terms read something beyond the first crossing that the four agreeing curves happen to share and the ramp does not.

We have submitted a diagnostic family that holds the crossing frame at 75 and varies only the shape before it, after it, and both, to separate these. The result will be reported in the final version; until it is, the honest statement of scope is that **the decomposition is established for step-like curves, the floor 0.35105 is measured on flat curves, and the crossing-frame model predicts step-like curves well and one graded curve poorly.** The headline result — a constant scoring 2.35291 — does not depend on any of this, because both constant_0.51 and never_crosses are flat curves and their difference is measured directly.

5.4 A property of the withheld ground truth, recovered

The linearity has a consequence the organisers did not intend to publish.

For a step at frame k , both timing terms equal $\max(t_{ai} - k, 0)$ on each contributing clip, so the score above the floor is $c \cdot E[\max(t_{ai} - k, 0)]$ with $c = w_{TTA} + w_{STTA}$. Differentiating, d/dk of that expectation is $-P(t_{ai} > k)$: **the slope at k measures the survival function of the accident-onset distribution, scaled by c .** The leaderboard is therefore reporting, one submission at a time, the shape of an annotation no entrant has seen.

Segment slopes, from consecutive pairs in the table above:

Interval	0–10	10–25	25–50	50–75	75–100	100–125	125–140
–slope	0.016667	0.016595	0.016658	0.016657	0.016649	0.012342	0.001908

Three things follow, stated with the bounds they actually support.

$c \approx 1/60$. The initial segment gives 0.016667, which is $1/60.00$ to five figures; since $P(t_{ai} > 0) = 1$, that segment estimates c directly.

Almost no onset before frame 100. The ratio of the 75–100 slope to the 0–10 slope is $0.016649/0.016667 = 0.99894$, so $P(t_{ai} \leq 100) \leq 0.11\%$ of contributing clips — about one or two clips in the private split, not zero, and we do not claim zero.

Almost none after frame 140. `step_at_140` and `never_crosses` agree at the displayed precision, so the excess $E[\max(t_{ai} - 140, 0)]$ is below $5 \times 10^{-6} / 0.016667 \approx 3 \times 10^{-4}$ frames. That bounds a *mean*, not a support: it permits a handful of late-onset clips while remaining invisible at five decimals.

Dividing the timing total by `c`:

The accident onset lies between frames 100 and 140 in all but a fraction of a percent of accident clips, with mean 120.1 (120.3 using the fitted rather than the initial slope) — that is, at about 4.0 s of a 5.00 s clip.

This is a measurement of hidden annotations obtained without them, and it explains mechanically why the constant does so well on this corpus: because the accident is always late in the window, an alarm at frame 0 is credited with roughly 120 frames — four seconds — of anticipation on every clip.

An unexplained residual. Under this model $|\text{slope}|$ must be non-increasing in `k`, because a survival function cannot rise. It is not: the 25–50 segment is steeper than the 10–25 segment by 6.3×10^{-5} , roughly ninety times the $\pm 6.7 \times 10^{-7}$ that five-decimal rounding permits over a fifteen-frame span. The straight line of \$5.3 is therefore very good but not exact, and the deviation is real. We suspect it is the same second-order effect that produces the `linear_ramp` discrepancy, and we report it rather than fit around it.

Two conventions this rests on, neither of them verified. When $k > t_{ai}$ the set $\{t_a : p_t > 0.5, t_a \leq t_{ai}\}$ is empty and the definition as published gives an undefined maximum; we assume the scorer takes it as zero. We also assume clips with no accident contribute nothing to the timing terms. Both assumptions are load-bearing for the $k > 100$ region and we flag them as such.

What we deliberately do not do is separate the weights from the metric values. All quantities above are products $w \cdot \text{metric}$. Note in particular that the ratio of the timing total to the slope equals the mean onset *conditional on the clips that contribute*, and that the positive rate cancels out of that ratio identically — so this measurement says nothing whatever about what fraction of the corpus contains an accident, and we make no such claim. The products are what determine the ranking, which is why reporting them is sufficient; separating a weight from a metric value would require the ground truth we correctly do not have.

5.5 Summary of all eighteen submissions

File	Public	Private
constant_0.51	2.35057	2.35291
constant_0.99	2.35057	2.35291
step_at_000	2.35057	2.35291
step_at_010	2.18391	2.18624
step_at_025	1.93391	1.93732
cross_then_dip	1.85057	1.85347
step_at_050	1.51724	1.52088

File	Public	Private
sigmoid_m0.40	1.35057	1.35428
cross_then_drop	1.22573	1.22698
step_at_075	1.10057	1.10445
sigmoid_m0.50	1.10057	1.10435
linear_ramp	1.06604	1.06725
sigmoid_m0.60	0.84398	0.84840
step_at_100	0.68391	0.68822
sigmoid_m0.70	0.55190	0.55763
step_at_125	0.37383	0.37967
step_at_140	0.35088	0.35105
never_crosses	0.35088	0.35105

Public and private scores agree to within 0.00584 across all eighteen, so the *blind* scores are stable across the split. We claim nothing from this about whether the teams' relative positions are split-stable, which we cannot observe.

6. Analysis: why the composition fails

Scope, fixed before the argument rather than after it: what follows is an analysis of a metric's algebra, illustrated by measurements on one benchmark with thirteen teams. Properties (a) and (b) are consequences of the definitions in §2.3 and hold wherever those definitions are used. Property (c) is a statement about what the algebra implies, not a survey finding. We have not audited the literature, and §8 states what that costs us.

The defect is not a badly chosen weight. It is a type error.

Write the score as $s = B + U$, where $B = w_{AP} \cdot AP + w_{AUC} \cdot AUC$ is bounded above by $w_{AP} + w_{AUC}$, and $U = w_{TTA} \cdot TTA + w_{STTA} \cdot STTA$ is bounded above by $(w_{TTA} + w_{STTA}) \cdot E[t_{ai}]$, a quantity that depends on the *corpus windowing*. Three properties follow.

(a) The unbounded term grows with clip length while the bounded one does not. $\max U / \max B$ scales linearly in $E[t_{ai}]$, so two research groups adopting the same weights on corpora windowed differently are not using the same metric, even though they report the same metric's name. On this benchmark $\max U$ is 2.00186, measured; $\max B$ is not identifiable from outside but is at least 0.49979, giving $\max U / \max B \leq 4.01$. Against B at chance the measured ratio is 5.70.

(b) The unbounded term has a constant as its global maximiser. Any *unconditioned* threshold-crossing earliness measure is maximised by crossing immediately and never returning; this holds for TTA and STTA as defined here. The maximiser reads no input, so the term it maximises can carry no information about the input. The qualifier matters and is the seed of the fix: an earliness measure that is *conditioned* — restricted to correctly classified

clips, or evaluated at a fixed operating point of the discrimination metric, or penalised by the false-alarm rate — is not maximised by a constant. §7.2 recommends exactly such a variant.

(c) Consequently the separating power of the score lies in B. To the extent that a submission crosses the threshold early and holds it, its U is at or near the maximum and it is separated from other such submissions only by B . We cannot verify this for any particular entry — we have no team’s curves but our own — so we state it as what the metric’s algebra implies for early-crossing submissions, not as an observation about anyone’s system.

Point (c) is why the failure is hard to notice from inside. A leaderboard of this shape can still order methods usefully, because B still varies. What is invisible without a control is that the *number* attached to each of them is dominated by a term available for free, so the reported score — the thing that goes into a paper’s table and gets compared across publications — is mostly a constant.

Two smaller mechanisms compound this, both visible in our own prior work:

Metric-enforcing post-processing. Any operation that clamps the risk score above the threshold and is followed by a report of “STTA compliance” has measured only that the clamp executed. Stated generally: a post-hoc operation that enforces a metric’s definition renders that metric vacuous. Our earlier pipeline did this, and reported 100% stable-anticipation compliance as a result.

Best-subset reporting. Selecting the best 50 of 1,417 clips and reporting their mean anticipation time is a statement about the selection, not the method. Our earlier draft did this too.

7. A corrected protocol

Each item is traced to a mechanism above. The first is the one we would keep if we could keep only one.

1. **Report a video-blind control with every anticipation result.** Compute the full metric on a constant above threshold, a linear ramp and a mid-clip step, and put those rows in the results table. This is a few minutes of work and it is diagnostic: if the control is within noise of the method, the metric is not measuring the method. We propose this become as reflexive as reporting a majority-class baseline in classification. (§5.1, §6b)
2. **Bound the earliness term, and condition it.** These are two separate repairs and both are needed. *Bounding* — normalising per clip so $TTA/t_{ai} \in [0, 1]$ — stops the term dominating the sum and stops the metric depending on how the corpus was windowed; it does **not** remove the constant as maximiser, which still scores 1 and still gets it for free. *Conditioning* is what removes the exploit: report earliness at a fixed operating point of the discrimination metric — mean TTA at 80% recall is the established form — so that a submission must first classify before its earliness is counted. If only one is adopted, adopt conditioning. (§6a, §6b)

3. **State the reference point for every timing metric.** t_{ai} or clip end. Reporting $(T - t_a) / \text{fps}$ under the name TTA inflates every result by a corpus-dependent constant. (§2.4)
4. **Report AP and AUC alongside any timing metric, never timing alone**, and state the positive:negative ratio. Without negatives there is no false-positive rate and any rising curve achieves perfect recall. (§6c)
5. **Report metrics with post-processing disabled as well as enabled.** If a number moves when a clamp is removed, the clamp was part of the measurement. (§6)
6. **Publish scores for a released control submission alongside the leaderboard.** Benchmark organisers can do in one afternoon what took us eighteen submissions: score a video-blind family and publish the numbers as a permanent floor on the leaderboard page. An entrant would then see at a glance how much of their score they earned.

A seventh applies specifically to competitions that withhold labels: **publish the metric weights**. Withholding labels is legitimate and prevents overfitting. Withholding the weights does not prevent the metric being exploited — this paper exploited it comprehensively without them, and eighteen submissions were enough to recover most of its structure — while it does prevent the one-line sanity check that would have surfaced the problem before the competition opened.

8. Limitations

One benchmark. We demonstrate on a single competition with 13 teams. The ranking result is therefore anecdotal in scale, and we do not present it as a finding about a field. The *decomposition*, however, is a property of the metric’s algebra: the fact that a threshold-crossing earliness term is maximised by a constant, and that summing it into a bounded term lets it dominate, holds independently of how many teams entered. What the competition supplies is a third-party scorer that let us measure the magnitude in a live setting rather than argue it in the abstract.

We cannot separate the weights from the metric values. All decomposed quantities are products $w \cdot \text{metric}$. Reporting them is sufficient for the argument — the products determine the ranking — but we cannot state, for example, what AP the corpus’s base rate implies.

We do not evaluate the corrected protocol on a labelled corpus. Items 2–5 of §7 are derived from the mechanisms, not demonstrated end-to-end, because this corpus publishes no labels. Demonstrating them requires a benchmark with released annotations (DAD, CCD or similar) and is the natural next step; we state it as future work rather than claim it here.

The control is a sufficient, not a necessary, condition. A metric on which a video-blind control scores poorly is not thereby sound. Our control detects this particular failure mode and makes no claim beyond it.

The caption leakage is unmeasured. We identify it (§3) and do not quantify it.

One discrepancy in the crossing-frame model is open. `linear_ramp` scores 0.037 below three curves that cross the threshold at the same frame (§5.3), and the segment slopes are not monotone as the model requires (§5.4). The diagnostics that separate the candidate explanations have been submitted and will be reported. Neither the headline result nor the timing total depends on the resolution — both are measured between flat curves — but the *split* of the timing total between TTA and STTA, and the interpretation of the fit as exact, do.

The character of the venue. This is a community competition with an undisclosed custom metric, not a long-running benchmark with a published scorer. That is what made the probe experiment possible; it also means we cannot inspect the scoring code, and every statement about the metric’s implementation in this paper is an inference from its outputs.

Our own system’s placement is not evidence about it. That our ensemble scored below the constant tells us about the metric, not about the ensemble; under a protocol that bounded the earliness term the comparison might go either way, and we do not know, because we cannot compute AP or AUC on this corpus.

9. Related work

[To be completed against the reference list. The sections to write: (i) supervised accident anticipation and the origin of the TTA-style metric; (ii) zero-shot and vision–language anticipation; (iii) benchmark-critique and shortcut-learning literature — Torralba & Efros on dataset bias, the Clever Hans literature, “Are we done with ImageNet?”, the reproducibility work in recommender systems where a similar “strong baselines beat the state of the art” finding was made, and the NLP hypothesis-only baselines for SNLI/MNLI, which is the closest methodological precedent: a control that reads only part of the input and scores far above chance.]

The nearest precedent is the hypothesis-only baseline in natural-language inference (Gururangan et al., 2018; Poliak et al., 2018): a model given only half of each input scored far above chance, demonstrating that the benchmark’s headline number was partly measuring something other than the task. Our control is more extreme — it reads none of the input — and the mechanism differs (metric composition rather than annotation artefacts), but the methodological lesson is the same, and its adoption in that field is the outcome we would like to see here.

10. Conclusion

A composite metric that adds an unbounded earliness term to bounded discrimination terms does not measure what its name suggests. On a live accident anticipation benchmark we measured the magnitude: the earliness terms are worth 5.70 times what the discrimination terms are worth to a blind submission, they are maximised exactly by a constant that reads no pixels, and that constant scores within 0.14874 of the winning entry and matches the eighth-placed team’s score at the precision the leaderboard displays. Using eighteen video-blind submissions as probes we recovered most of the metric’s functional form from outside the competition — predicting held-out step-like curves to within 6.4×10^{-3} , failing on one graded

curve in a way we report rather than smooth — and along the way recovered the support and mean of the withheld accident-onset distribution.

The remedy is cheap. Bound the earliness term, or report it separately. And run the control: a constant costs one submission and settles in an afternoon a question that otherwise propagates through a literature.

We report our own entry, which scored below the constant, as the case study.

Reproducibility

All code, all submission files, and the complete score log will be released on publication. (*Repository URL withheld here for double-blind review; it is supplied to the editor.*) The central result is one line:

```
risk = [0.51] * 150
```

Acknowledgements

[To be added — including the AUTOPILOT organisers, whose decision to leave late submission open is what made the probe experiment possible.]

References

[Existing reference list to be carried over from the prior manuscript, plus the benchmark-critique literature listed in §9.]

Benchmark citation, as requested by the organisers:

AUTOPILOT-COG. *Zero-shot Accident Anticipation*.
<https://kaggle.com/competitions/zero-shot-taa>, 2026. Kaggle.